Lab 4: d-separation

Department of Artificial Intelligence, SVNIT Surat · B.Tech. AI, Semester VII

Write a program that reads a directed acyclic graph and a list of queries from a file, and prints for each query whether the two nodes are d-separated by the given evidence set.

Use the trail-based criterion from the lectures: `x` and `y` are d-separated by `Z` when every trail between them is blocked.

Input format

Nodes are numbered `0` to `n-1`. Every record is on its own line; a line is one record, so read the file line by line.

```
n m                 n nodes, m edges
u v                 m lines, one per directed edge u -> v
q                   number of queries
x y z1 z2 ... zk    q lines. The first two numbers are the pair.
                    Everything after them on that line is the
                    evidence set Z, which may be empty.
```

In every query `x`, `y` and the nodes of `Z` are distinct: `x` is never `y`, and neither `x` nor `y` ever appears in `Z`.

Output

One line per query, **in the same order as the queries appear in the input**. Print `YES` if `x` and `y` are d-separated by `Z`, otherwise `NO`. So `q` queries produce exactly `q` lines.

Print exactly `YES` or `NO` in capitals, nothing else on the line. No headings, no query numbers, no blank lines.

Example

`case_p4_sample.in` is the Student network from the lectures, with 0 = Difficulty, 1 = Grade, 2 = Intelligence, 3 = Letter, 4 = SAT. Edges $D \to G$, $I \to G$, $I \to S$, $G \to L$.

The file in full, followed by the output your program should produce for it. These are two separate files.

```
case_p4_sample.in
-----------------
5 4
0 1
2 1
2 4
1 3
12
0 2
0 2 1
0 2 3
0 4
0 4 1
0 4 2
0 4 1 2
2 3 1
3 4
3 4 2
1 4 2
0 3
```

```
expected output
---------------
YES
NO
NO
YES
NO
YES
YES
YES
NO
YES
YES
NO
```

Run and submit

Your program must take the input filename as its first command-line argument and write the answers to standard output:

```
python3 lab4.py case_p4_main.in
```

It will be run on an input file you have not seen, which may be in a different directory and may have a different number of nodes and queries. Do not hard-code the name of any file, or the size of anything.

Run on `case_p4_sample.in` first. All twelve answers must match the ones shown above; they are small enough to work out on paper as well.

Then run on `case_p4_main.in`.

Submit exactly two files, with these names:

```
lab4.py            all of your code, in this one file
lab4_output.txt    the lines your program printed for case_p4_main.in
```

Nothing else. No second `.py` file, no zip archive, no notebook.

Standard library only. `networkx` and `pgmpy` implement d-separation directly and are not permitted.

The Bayes-Ball / reachability algorithm answers the same question without enumerating trails; do not use it here. It is the subject of a later lab.